

TASK 1 (A):

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May 5 20:27
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/...
~/Desktop/lab 9 os
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ cat tia.c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

int icecreamRemaining = 10;
sem_t salesman;

void* eatIcecream(void* arg) {
    int id = *((int*)arg);

    while (1) {
        sem_wait(&salesman);

        if (icecreamRemaining <= 0) {
            sem_post(&salesman);
            break;
        }

        printf("Person %d is taking ice cream. Remaining = %d\n", id, icecreamRemaining);
        icecreamRemaining--;

        sem_post(&salesman);
        sleep(1);
    }

    printf("Person %d done eating.\n", id);
    pthread_exit(NULL);
}

int main() {
    pthread_t t1, t2, t3;
    int ids[3] = {1, 2, 3};

    sem_init(&salesman, 0, 1);

    pthread_create(&t1, NULL, eatIcecream, &ids[0]);
    pthread_create(&t2, NULL, eatIcecream, &ids[1]);
    pthread_create(&t3, NULL, eatIcecream, &ids[2]);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    pthread_join(t3, NULL);

    sem_destroy(&salesman);
    return 0;
}

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ gcc tia.c -o exe -lpthread
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ ./exe
Person 1 is taking ice cream. Remaining = 10
Person 3 is taking ice cream. Remaining = 9
Person 2 is taking ice cream. Remaining = 8
Person 1 is taking ice cream. Remaining = 7
Person 3 is taking ice cream. Remaining = 6
Person 2 is taking ice cream. Remaining = 5
Person 1 is taking ice cream. Remaining = 4
Person 3 is taking ice cream. Remaining = 3
Person 2 is taking ice cream. Remaining = 2
Person 1 is taking ice cream. Remaining = 1
Person 3 done eating.
Person 2 done eating.
Person 1 done eating.
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$
```

TASK 1 (B):

```
May 5 20:30
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/...
~/Desktop/lab 9 os
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ cat tib.c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#include <stdlib.h>
int icecreamRemaining = 10;
sem_t salesman;
void* eatIcecream(void* arg) {
    int id = *((int*)arg);

    while (1) {
        sem_wait(&salesman);

        if (icecreamRemaining <= 0) {
            sem_post(&salesman);
            break;
        }

        printf("Person %d buying ice cream. Remaining = %d\n", id, icecreamRemaining);
        icecreamRemaining--;

        sem_post(&salesman);
        printf("Person %d counting money\n", id);
        sleep(1 + rand() % 2);
    }

    printf("Person %d finished.\n", id);
    pthread_exit(NULL);
}

int main() {
    pthread_t t1, t2, t3;
    int ids[3] = {1, 2, 3};

    sem_init(&salesman, 0, 1);

    pthread_create(&t1, NULL, eatIcecream, &ids[0]);
    pthread_create(&t2, NULL, eatIcecream, &ids[1]);
    pthread_create(&t3, NULL, eatIcecream, &ids[2]);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    pthread_join(t3, NULL);

    sem_destroy(&salesman);
    return 0;
}

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ gcc tib.c -o exe -lpthread
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ ./exe
Person 1 buying ice cream. Remaining = 10
Person 1 counting money
Person 3 buying ice cream. Remaining = 9
Person 3 counting money
Person 2 buying ice cream. Remaining = 8
Person 2 counting money
Person 3 buying ice cream. Remaining = 7
Person 3 counting money
Person 2 buying ice cream. Remaining = 6
Person 2 counting money
Person 1 buying ice cream. Remaining = 5
Person 1 counting money
Person 3 buying ice cream. Remaining = 4
Person 3 counting money
Person 3 buying ice cream. Remaining = 3
Person 3 counting money
Person 1 buying ice cream. Remaining = 2
Person 1 counting money
Person 2 buying ice cream. Remaining = 1
Person 2 counting money
Person 3 finished.
Person 2 finished.
Person 1 finished.
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$
```

TASK 2:

```
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/...  
~/Desktop/lab 9 os  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ cat t2.c  
#include <stdio.h>  
#include <pthread.h>  
#include <unistd.h>  
  
#define CUSTOMERS 10  
  
sem_t weighingCounter;  
sem_t securityCounter;  
sem_t boardingCounter;  
  
void* passenger(void* arg){  
    int id = *((int*)arg);  
  
    printf("Passenger %d arrived\n", id);  
    sem_wait(&weighingCounter);  
    printf("Passenger %d weighing luggage.\n", id);  
    sleep(4);  
    sem_post(&weighingCounter);  
  
    sem_wait(&securityCounter);  
    printf("Passenger %d in security check...\n", id);  
    sleep(7);  
    sem_post(&securityCounter);  
  
    sem_wait(&boardingCounter);  
    printf("Passenger %d getting boarding pass.\n", id);  
    sleep(3);  
    sem_post(&boardingCounter);  
  
    printf("Passenger %d done boarding process\n", id);  
    pthread_exit(NULL);  
}  
  
int main(){  
    pthread_t t[CUSTOMERS];  
    int ids[CUSTOMERS];  
  
    sem_init(&weighingCounter, 0, 2);  
    sem_init(&securityCounter, 0, 2);  
    sem_init(&boardingCounter, 0, 2);  
  
    for (int i = 0; i < CUSTOMERS; i++){  
        ids[i] = i + 1;  
        pthread_create(&t[i], NULL, passenger, &ids[i]);  
    }  
  
    for (int i = 0; i < CUSTOMERS; i++){  
        pthread_join(t[i], NULL);  
    }  
    return 0;  
}  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$  
  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ gcc t2.c -o exe -lpthread  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ ./exe  
Passenger 2 arrived  
Passenger 3 weighing luggage.  
Passenger 9 arrived  
Passenger 9 weighing luggage.  
Passenger 8 arrived  
Passenger 4 arrived  
Passenger 6 arrived  
Passenger 5 arrived  
Passenger 7 arrived  
Passenger 1 arrived  
Passenger 10 arrived  
Passenger 2 arrived  
Passenger 9 in security check...  
Passenger 8 weighing luggage.  
Passenger 3 in security check...  
Passenger 4 weighing luggage.  
Passenger 6 weighing luggage.  
Passenger 5 weighing luggage.  
Passenger 9 getting boarding pass.  
Passenger 8 in security check...  
Passenger 3 getting boarding pass.  
Passenger 4 in security check...  
Passenger 7 weighing luggage.  
Passenger 1 weighing luggage.  
Passenger 9 done boarding process  
Passenger 3 done boarding process  
Passenger 2 weighing luggage.  
Passenger 10 weighing luggage.  
Passenger 8 getting boarding pass.  
Passenger 6 in security check...  
Passenger 4 getting boarding pass.  
Passenger 5 in security check...  
Passenger 8 done boarding process  
Passenger 4 done boarding process  
Passenger 6 getting boarding pass.  
Passenger 7 in security check...  
Passenger 5 getting boarding pass.  
Passenger 1 in security check...  
Passenger 6 done boarding process  
Passenger 5 done boarding process  
Passenger 7 getting boarding pass.  
Passenger 2 in security check...  
Passenger 1 getting boarding pass.  
Passenger 10 in security check...  
Passenger 7 done boarding process  
Passenger 1 done boarding process  
Passenger 3 getting boarding pass.  
Passenger 10 getting boarding pass.  
Passenger 2 done boarding process  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$
```

TASK 3:

```
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/...  
~/Desktop/lab 9 os  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ cat t3.c  
#include <stdio.h>  
#include <pthread.h>  
  
int counter=0;  
pthread_mutex_t lock;  
  
void* increment(void* arg){  
    for (int i = 0; i < 1000; i++){  
        pthread_mutex_lock(&lock);  
        counter++;  
        pthread_mutex_unlock(&lock);  
    }  
    pthread_exit(NULL);  
}  
  
int main(){  
    pthread_t t1, t2;  
  
    pthread_mutex_init(&lock, NULL);  
  
    pthread_create(&t1, NULL, increment, NULL);  
    pthread_create(&t2, NULL, increment, NULL);  
  
    pthread_join(t1, NULL);  
    pthread_join(t2, NULL);  
  
    printf("Final Counter Value = %d\n", counter);  
  
    pthread_mutex_destroy(&lock);  
    return 0;  
}  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$  
  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ gcc t3.c -o exe -lpthread  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$ ./exe  
Final Counter Value = 2000  
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop/lab 9 os$
```

